IT4010 Research Project — 2026 June

Mid month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	1st June – 15th June 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	60%

Section C — Tasks This Period

C1. Tasks Completed

- Reviewed the Raspberry Pi edge-gateway plan agreed after PP1 and defined the gateway's responsibilities: local MQTT ingestion, payload validation, evidence storage, bin prioritisation and bounded feedback to the nodes.
- Decided to keep the gateway on the laptop for the prototype. It runs the same Node.js and Mosquitto stack a Raspberry Pi would run, so the gateway can move to a Pi later without redesign.
- Restructured the backend so the gateway logic sits in its own module (gateway.mjs), separate from the REST API and the WebSocket server.
- Set up a basic Expo / React Native mobile shell that reads bin status from the backend API (/api/readings).

C2. Tasks Planned for Next Period

- Bring up a second, low-cost bin node on the ESP8266 (ESP-12E) running the same four-state policy.
- Make sensor reads bounded in time so a slow or failed DHT22 read cannot stall the 1 s policy loop.
- Continue the bi-weekly logbook entries and prepare for the next supervisor review.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- A Raspberry Pi gateway would add setup and debugging time without changing the research contribution. Mitigation: run identical gateway software on the laptop and keep the Pi as the deployment target.
- The full mobile scope (schedules, alerts, issue reporting) is large. Kept to a basic status shell so effort stays on the core policy and its evaluation.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	1st June – 15th June 2026
Page / Entry Numbers	

Student Declaration
I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement
I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 June

End of Month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	16th June – 30th June 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	63%

Section C — Tasks This Period

C1. Tasks Completed

- Brought up the second node, bin02, on an ESP8266 (ESP-12E NodeMCU) with HC-SR04 fill and DHT22 sensing. This low-cost node class has no gas sensor.
- Ported the four-state policy (STABLE, ACTIVE, ALERT, CONSERVE), the per-state AoI floors, the dynamic deadband and AIMD to the ESP8266 build, so both node classes run one policy.
- Wrote a bounded-time DHT22 driver (BoundedDHT) so a failed temperature or humidity read returns within a fixed time instead of blocking the policy loop.
- Vendored the bin02 libraries (PubSubClient, ArduinoJson, BoundedDHT) into the project so the firmware builds reproducibly offline.

C2. Tasks Planned for Next Period

- Harden the gateway: validate topic, node identity, value ranges and payload size before storing a reading.
- Add a priority ranking of bins at the gateway.
- Test two-node ingestion end to end.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- The ESP8266 has less memory and a different Wi-Fi stack than the ESP32. Kept the payload compact and confirmed both builds fit.
- ESP8266 boot-strapping pins (GPIO 0, 2, 15) must stay free. Used TRIG 13 / ECHO 14 for the HC-SR04 and GPIO 4 for the DHT22.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	16th June – 30th June 2026
Page / Entry Numbers	

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 July

Mid month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	1st July – 15th July 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	67%

Section C — Tasks This Period

C1. Tasks Completed

- Implemented gateway validation: topic format, known node identity, value ranges and payload size are checked before a reading is stored.
- Extended the SQLite evidence store and the REST API for multiple nodes, with a sensor-profile field per node.
- Implemented bin prioritisation at the gateway: bins are ranked by state first, then by fill level and data freshness.
- Tested simultaneous telemetry from bin01 (ESP32) and bin02 (ESP8266) through the local Mosquitto broker.

C2. Tasks Planned for Next Period

- Add bounded gateway feedback to the nodes: a time-limited "report faster" lease with acknowledgement.
- Add network-health monitoring so feedback is only requested when the link is healthy.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- The two node classes report different sensor sets. Handled with a sensor-profile field so the gateway and dashboard know which sensors each node has.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	1st July – 15th July 2026
Page / Entry Numbers	

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 July

End of Month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	16th July – 31st July 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	72%

Section C — Tasks This Period

C1. Tasks Completed

- Implemented bounded gateway feedback: the gateway can ask a node to report at a shorter interval for a limited time (TTL). Every command carries a request id and is acknowledged by the node.
- Made sure the node's own urgency and energy rules always take priority over a gateway request.
- Implemented a gateway network monitor that samples latency, packet loss, Wi-Fi signal and MQTT status and combines them into a link-health score. Optional feedback requests are allowed only when the link is healthy.
- Stored the network samples with run labels for later analysis.

C2. Tasks Planned for Next Period

- Extend the dashboard with a decision trace and per-sensor status.
- Prepare unattended evaluation runs and the Progress Presentation 2 material.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- Gateway commands could conflict with the node's energy policy. Resolved by making leases time-limited and letting node-side rules win.
- The network monitor measures the gateway's own link, not each node's radio link. Recorded as a limitation; per-node RSSI is future work.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	16th July – 31st July 2026
Page / Entry Numbers	

Student Declaration
I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement
I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 August

Mid month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	1st August – 15th August 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	78%

Section C — Tasks This Period

C1. Tasks Completed

- Extended the React dashboard: per-bin cards with sensor status and fault flags, and a decision trace showing the state, the trigger and whether the node transmitted.
- Added the multi-node view so bin01 and bin02 appear side by side with their own state and freshness.
- Ran resilience tests: broker stopped and restarted, MQTT reconnect behaviour, and node behaviour with no broker available.
- Prepared an unattended overnight run script that switches BASELINE and ADAPTIVE phases through the backend API.

C2. Tasks Planned for Next Period

- Run the unattended overnight BASELINE vs ADAPTIVE comparison.
- Prepare the PP2 deck and a Q&A guide.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- Reconnect tests showed the node must keep sensing and deciding while the broker is down. Confirmed the policy loop continues and reconnect is retried without blocking it.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	1st August – 15th August 2026
Page / Entry Numbers	

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 August

End of Month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	16th August – 31st August 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation
Overall Status	On Track
% Completion (Est.)	88%

Section C — Tasks This Period

C1. Tasks Completed

- Ran an unattended overnight BASELINE vs ADAPTIVE run (30 August). It exposed bursts of change-based transmissions caused by movement in front of the ultrasonic sensor.
- Fixed five defects found during the runs (29 to 30 August): a failed HC-SR04 read was treated as 0% fill and fired false ALERTs (it now holds the last good value and flags sensor_fault); AIMD stepped once per loop tick instead of once per transmission; fill-rate units were mixed (now %/hr over a 60 s window, clamped at ±500); the device tx_count reset on reboot (totals now come from stored rows); and readings had no server time (every row now carries rx_ts and a run_label).
- Ran diagnostic runs on 31 August (firmware repair, router switch, gateway hardware) to separate firmware, network and gateway causes.
- Mid-year grade released (Average). Reviewed it and decided to prioritise measured evidence over new features.
- Prepared the PP2 deck (15 slides) and a PP2 Q&A guide.

C2. Tasks Planned for Next Period

- Deliver Progress Presentation 2.
- Stabilise sensing: remove MQ-5 from the decision path and verify the DHT22.

- Install INA219 power measurement and run matched BASELINE vs ADAPTIVE trials.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- The overnight bursts mean that run's transmission counts cannot be used as a TCR result. Treated as diagnostic evidence only; matched trials planned instead.
- Several defects only appeared in long unattended runs. Every row now carries a server timestamp and run label so runs can be analysed reliably.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	16th August – 31st August 2026
Page / Entry Numbers	

Student Declaration

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Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 September

Mid month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	1st September – 15th September 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Implementation and Evaluation
Overall Status	On Track
% Completion (Est.)	90%

Section C — Tasks This Period

C1. Tasks Completed

- Delivered Progress Presentation 2 on 1 September with a live two-node demonstration (run label pp2-live-20260901): on-device policy, laptop gateway, evidence store and dashboard.
- Ran a bin02 diagnostic with no sensors attached (1 September) to separate node faults from gateway faults.
- Collected gateway network-health samples from 1 to 3 September (about 7,900 samples).
- Started post-PP2 stabilisation: took MQ-5 out of the decision path (ENABLE_MQ5 = 0) because its readings were unstable; MQ-135 stays as a raw, uncalibrated gas channel.

C2. Tasks Planned for Next Period

- Update the literature review with 2022 to 2026 work and re-check the research gap.
- Integrate INA219 power measurement on both nodes.
- Plan the research paper and the research website.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- PP2 showed the implementation works but the energy claim still needs measurement. Plan: INA219 on each node's power input and matched runs on the same hardware.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	1st September – 15th September 2026
Page / Entry Numbers	

Student Declaration

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Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 September

End of Month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	16th September – 30th September 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Evaluation
Overall Status	On Track
% Completion (Est.)	92%

Section C — Tasks This Period

C1. Tasks Completed

- Completed literature review v4 (28 September): a wider 2022 to 2026 search; Wu et al. (Sensors 2026) identified as the closest smart-bin work; Gap 1 restated as an integration contribution; an 11-system comparison matrix.
- Selected and scored five papers for the supervisor's paper tracker (Giordano 2023, Suryavansh 2021, Delfani 2024, Wu 2026, Addas 2024).
- Planned the INA219 experiment: sensor on the node's 5 V input, BASELINE and ADAPTIVE runs of equal length on the same node.

C2. Tasks Planned for Next Period

- Integrate the INA219 on both nodes.
- Draft the research paper.
- Build the research website to the department guide.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- Wu et al. (2026) narrows the gap. Mitigation: state the contribution as the joint four-output policy on a bin node, not as a new mechanism, and cite Wu, Giordano and Suryavansh directly.
- Some sources could not be opened in full. Flagged as open items and not used for claims.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	16th September – 30th September 2026
Page / Entry Numbers	

Student Declaration
I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement
I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 October

Mid month Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	1st October – 15th October 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Evaluation
Overall Status	On Track
% Completion (Est.)	95%

Section C — Tasks This Period

C1. Tasks Completed

- Integrated INA219 voltage and current measurement on both nodes, on the 5 V input, with the readings included in each payload.
- Prepared the research paper draft (14 October, 11 pages).
- Planned the research website to the department guide: Home, Domain, Milestones, Documents, Presentations, About Us and Contact Us.

C2. Tasks Planned for Next Period

- Verify both nodes sensor by sensor.
- Build the research website.
- Run the matched evaluation and prepare the final presentation.

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- The INA219 on the 5 V input includes the MCU and the MQ-135 heater, so absolute energy differs between node classes. Each node is compared only with itself.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	1st October – 15th October 2026
Page / Entry Numbers	

Student Declaration
I confirm that this log entry is a true record of my research progress.

Signature: _____
Name: Perera G.M.T
Date: _____

Supervisor Acknowledgement
I confirm I have reviewed this bi-weekly progress submission.

Signature: _____
Name: Mr. Dinith Primal
Date: _____

IT4010 Research Project — 2026 October

Final Progress Log Entry

Section A — Student Information

Student Name	Perera GMT
Registration No.	IT21196638
Project Title	Edge-based energy efficient IoT driven smart waste management system
Supervisor	Mr. Dinith Primal
Co-Supervisor	Ms. Akshi De Silva
Research Cluster	Software Systems & Technologies (SST)
Specialization	Information Technology
Submission Period	16th October – 20th October 2026
Date Submitted	

Section B — Progress Summary

Current Project Stage	Final evaluation and presentation
Overall Status	On Track
% Completion (Est.)	98%

Section C — Tasks This Period

C1. Tasks Completed

- 17 October: verified both nodes sensor by sensor with probe sketches and serial capture, recorded pins, MAC addresses and measured values (HARDWARE.md), backed up the evidence database and upgraded the database driver for Node.js 24.
- 19 October: built the research website (seven pages, HTML and CSS) from a small generator script.
- 19 to 20 October: moved development to a dedicated laptop with the old laptop as the edge gateway, cleaned the repository (build output and databases untracked, secrets kept out) and committed a snapshot.
- Added a state-change transmission so CONSERVE and STABLE changes reach the dashboard within about 10 s.
- Found and fixed defects in the evaluation scripts (alert-recall label, mixed nodes, event time zone, phase length) and checked the fixes on recorded data.
- Served the dashboard from the edge gateway to any device on the site network, and resolved an MQTT connection problem between bin01 and the gateway.
- Measured the real bin depth and set it per node, flashed bin01, and ran the matched evaluation on 20 October: quiet BASELINE and ADAPTIVE runs for transmission reduction and energy, and an alert run for alert recall.

- Prepared the final presentation, including a commercialisation plan.

C2. Tasks Planned for Next Period

- Final presentation, viva and website evaluation (20 October).
- Submit the research paper (23 October) and the final thesis (28 October).
- Paper publication evidence (1 December).

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

- The bins are smaller than the 40 cm the firmware assumed, which would have read an empty bin as about 58% full. Fixed by measuring each bin and setting the depth per node.
- Three alert events in a short run lower the measured transmission reduction. Reported two figures: a steady-state run and an alert stress run.

Section E — Supervisor Meeting Notes

Date of Meeting	
Mode	

Key Points Discussed

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Section F — Research Logbook Reference

Logbook Entry Dates	16th October – 20th October 2026
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